Faster scalar multiplication on elliptic curves

Nicolas Sarkis
Advisors: Razvan Barbulescu and Damien Robert

Institut de Mathématiques de Bordeaux

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Outline

1. Elliptic curves
   - Cryptography
   - Elliptic curves
2. Kummer lines arithmetic
   - Montgomery xz-coordinates
   - Theta coordinates
   - Hybrid ladder

Elliptic curves

Symmetric cryptography (1/2)

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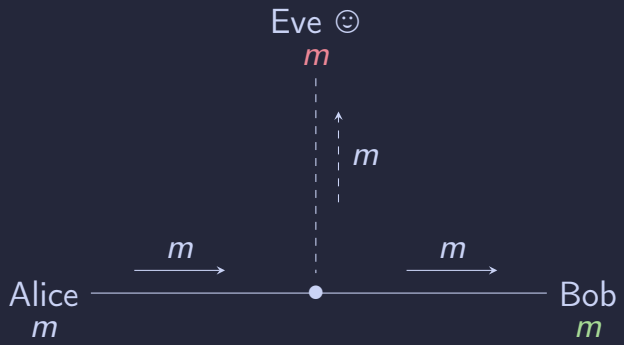

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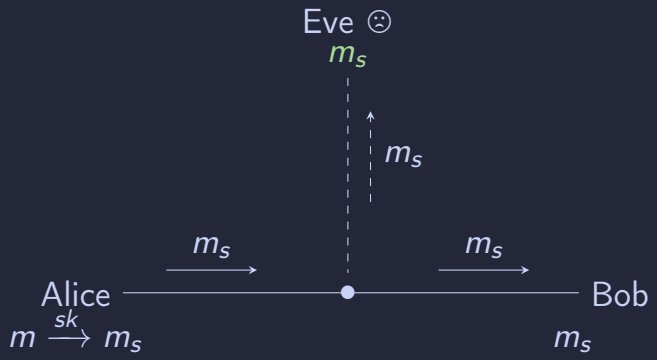

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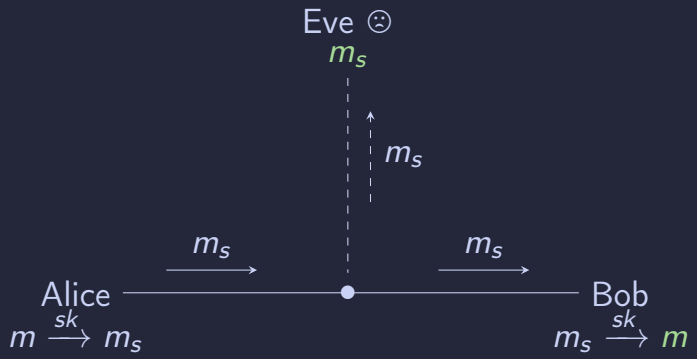

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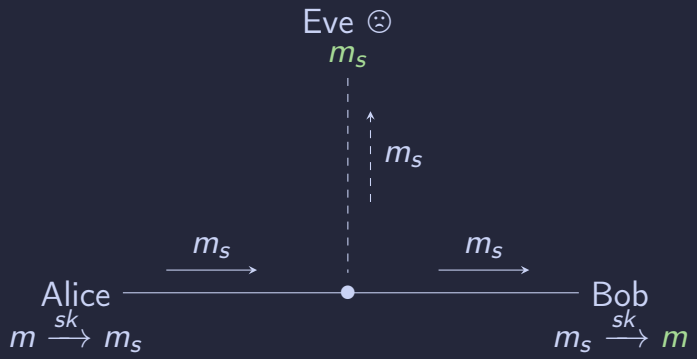

How do they get the same key?

Asymmetric cryptography: Diffie-Hellman key exchange

- Public: G is a finite cyclic group of order n, $g \in G$ is a generator: $G = \langle g \rangle$.
- Alice chooses a secret number $0 < a < n$, and Bob a secret number $0 < b < n$.

$$G =< g >$$

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a, g	b, g

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How to choose a group G? (historically, $G = \mathbb{F}_p^\times$)

Elliptic curves (char $k \neq 2, 3$)

- Short Weierstrass (general case):

$$E : y^2 = x^3 + ax + b$$

- Montgomery curves (less common):

$$E : By^2 = x(x^2 + \mathcal{A}x + 1)$$

- Twisted Edwards curves (isomorphic to Montgomery):

$$E : ax^2 + y^2 = 1 + dx^2y^2$$

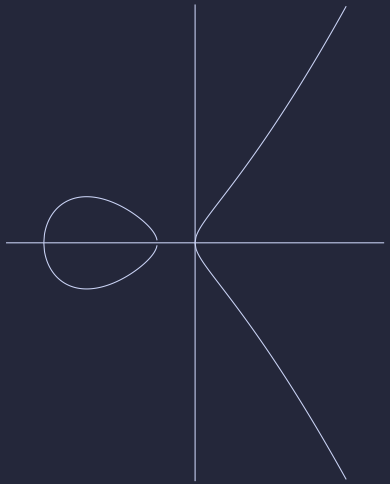

Figure: A Montgomery curve

Geometric addition law

Neutral element: point at infinity $\mathcal{O} = (0 : 1 : 0)$.

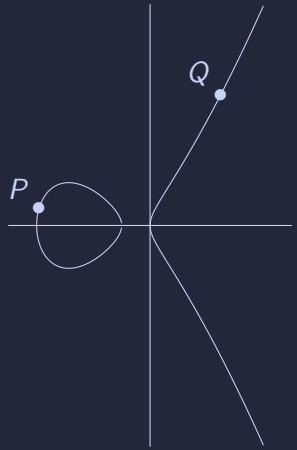

Figure: $P + Q$ with $P \neq Q$

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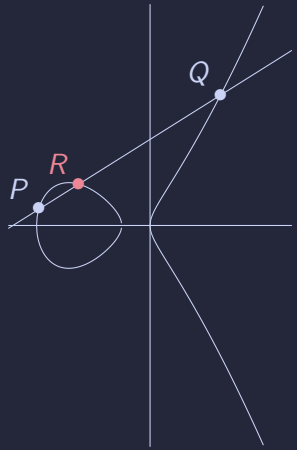

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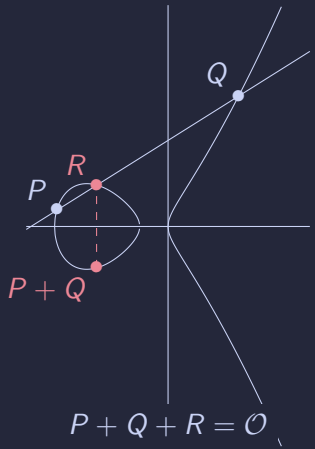

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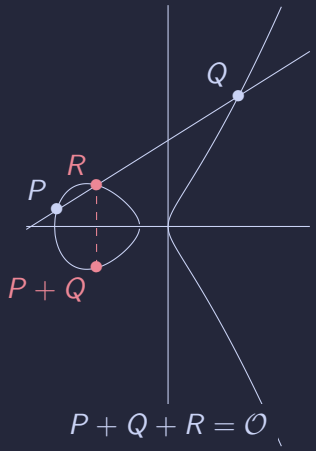

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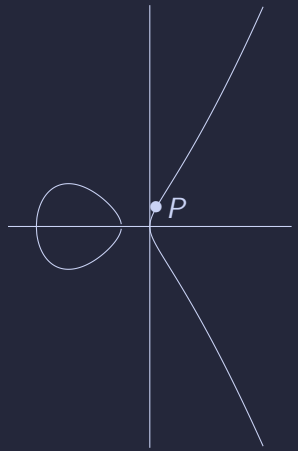

Figure: $2 \cdot P$

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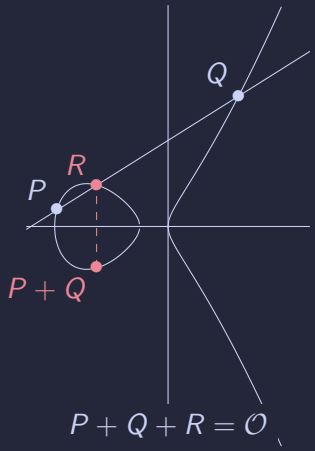

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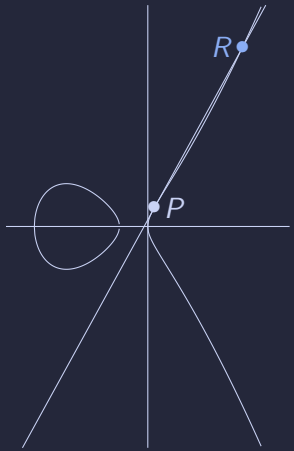

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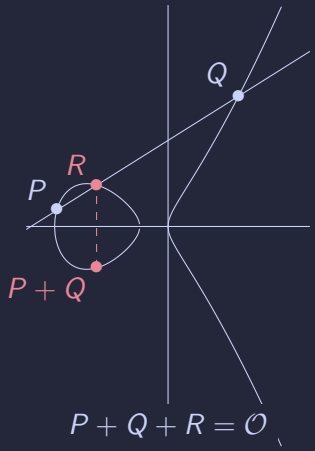

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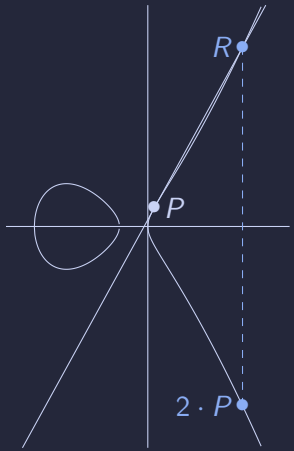

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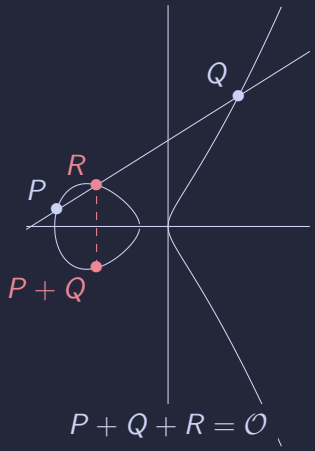

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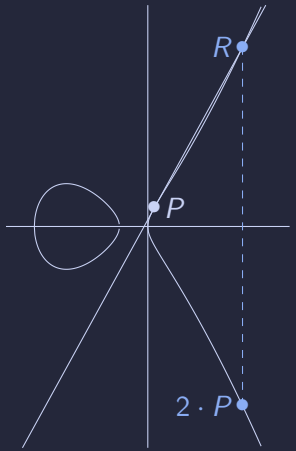

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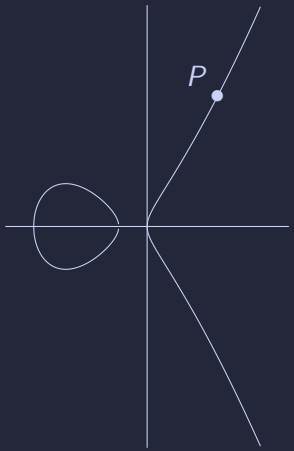

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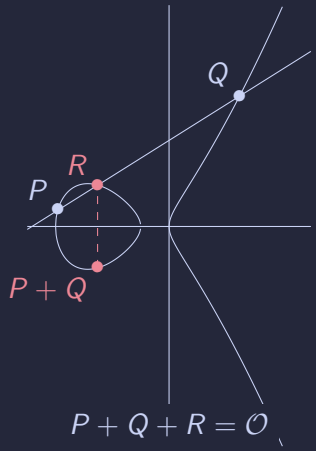

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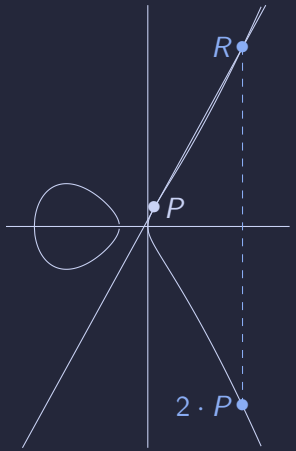

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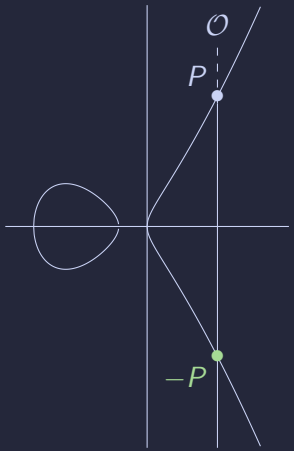

Figure: $-P$

Analytic addition law from geometric law

Short Weierstrass curve:

$$E : y^2 = x^3 + ax + b$$

$P = (x_1, y_1)$ and $Q = (x_2, y_2)$ points on E.

- If $P = -Q$, then $P + Q = \mathcal{O}$.
- If $P = \mathcal{O}$, then $P + Q = Q$.
- Else, $P + Q = (x_3, y_3)$ with $x_3 = \lambda^2 - x_1 - x_2$ and $y_3 = \lambda(x_1 - x_3) - y_1$.
  - If $P \neq Q$:
  $$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$$
  - Otherwise, $P = Q$ and:
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Computation-wise, on finite field, we don't really appreciate quotients...

Projective coordinates

Add a variable to store quotients:

$$P = (x, y) = (X : Y : Z) \text{ where } x = \frac{X}{Z} \text{ and } y = \frac{Y}{Z}$$

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- M, S: multiplication/square of elements of k.
- m_0: multiplication of an element of k by a curve constant.

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Addition

Achieved in $12M + 5m_0$.

Doubling

Achieved in $5M + 6S + 1m_0$.

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Twisted Edwards: $ax^2 + y^2 = 1 + dx^2y^2$

Addition

Achieved in $10M + 1S + 2m_0$.

Doubling

Achieved in $3M + 4S + 1m_0$.

Why counting M, S and m_0 differently?

Squaring S (in $\mathbb{F}_{p^2} = \mathbb{F}_p[i]$ with $i^2 = -1$ when $p \equiv 3 \bmod 4$)

M $(a+ib)(c+id) = ac - bd + i(ac + bd - (a-b)(c-d))$ $(3M(\mathbb{F}_p))$

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So $S/M \approx 2/3$, it is better to have squares.

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Small constant m_0

Represent numbers as polynomials in $r = 2^{64}$:

$$n, m = P(r), Q(r)$$

Then $nm = PQ(r)$, not linear in general.
But if P is monomial (e.g. n fits in 1 computer word), then nm is linear.

Torsion points

Definition

A m-torsion point P on an elliptic curve E is a point such that $m \cdot P = \mathcal{O}$. The group of m-torsion points is denoted $E[m]$.

If char $k = 0$ or $p > 0$ and $m \nmid p$, then $E[m] \simeq (\mathbb{Z}/m\mathbb{Z})^2$

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Example

Set $E : y^2 = x^3 + ax + b$ and $P = (x, y)$ a point of 2-torsion on E.
Then $P = -P$ i.e. $(x, y) = (x, -y)$ so $y = 0$ and $x^3 + ax + b = 0$.

A Montgomery curve always have at least one 2-torsion point $(0 : 0 : 1)$ (not to be mistaken with $\mathcal{O} = (0 : 1 : 0)$):

$$M : By^2 = x(x^2 + \mathcal{A}x + 1)$$

About rational points

All properties are given over an algebraically closed field, as well as the definition.

$$E : y^2 = x(x^2 + x + 1)$$

- E is well-defined over $\mathbb{Q}$ but $x(x^2 + x + 1) = 0$ only has one rational root.

$$E(\mathbb{Q})[2] = \langle (0,0) \rangle \subsetneq E(\overline{\mathbb{Q}})[2] = E[2] = \langle (0,0), (j,0) \rangle$$

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  $j^2 + j + 1 = 0$ and $(j,0) \notin E(\mathbb{Q})$.
- E is well-defined over $\mathbb{F}_7$ and $x(x^2 + x + 1) = x(x-2)(x-4)$:

$$E(\mathbb{F}_7)[2] = \langle (0,0), (2,0) \rangle = E(\overline{\mathbb{F}_7})[2] = E[2]$$

  Then $E[2] = \{\mathcal{O}, (0,0), (2,0), (4,0)\}$ and $(4,0) = (0,0) + (2,0)$.

Kummer lines arithmetic

Kummer lines

Focus on Montgomery curves:

$$E : By^2 = x(x^2 + \mathcal{A}x + 1)$$

If $P = (x : y : z)$, then $-P = (x : -y : z)$.
If one forgets about the sign of y, we get a Kummer line $\mathcal{K} = E/\{\pm 1\}$.

Montgomery xz-coordinates

This map is a degree 2 covering ($\pi^{-1}(x : z) = \{(x : \pm y : z)\}$ except if $y = 0$):

$$E \xrightarrow{\pi} \mathbb{P}^1$$

$$(x : y : z) \mapsto \begin{cases} \infty := (1 : 0) & \text{if } (x : y : z) = \mathcal{O} \\ \frac{x}{z} := (x : z) & \text{otherwise} \end{cases}$$

Notation: $\pi(P) = [P]$.

What about the addition law?

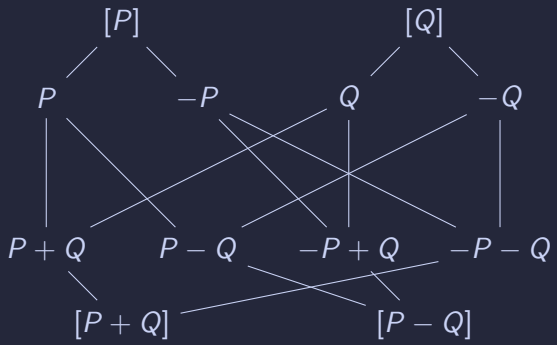

Figure: Two possible choices

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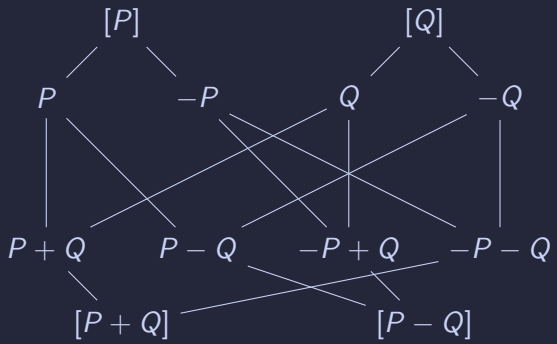

Figure: Two possible choices

However, if we know $[P]$, $[Q]$, $[P-Q]$, we can compute $[P+Q]$.

2- and 4-torsion behaviour on the Kummer line

Let $E : y^2 = x(x^2 + \mathcal{A}x + 1)$, $T \in E$ a 2-torsion point and $P \in E$.

2-torsion

Because $T = -T$, on the Kummer line $[P + T] = [P - T]$.
$\rightsquigarrow \tau_T$, the translation by T is well-defined on the Kummer line.
If $T = (0, 0) = (0 : 0 : 1)$, then $[T] = (0 : 1)$ and $\tau_T : (x : z) \mapsto (z : x)$.

2- and 4-torsion behaviour on the Kummer line

Let $E : y^2 = x(x^2 + \mathcal{A}x + 1)$, $T \in E$ a 2-torsion point and $P \in E$.

2-torsion

Because $T = -T$, on the Kummer line $[P + T] = [P - T]$.
$\rightsquigarrow \tau_T$, the translation by T is well-defined on the Kummer line.
If $T = (0, 0) = (0 : 0 : 1)$, then $[T] = (0 : 1)$ and $\tau_T : (x : z) \mapsto (z : x)$.

Assume we have T' a 4-torsion point above $T = (0 : 0 : 1)$ $(2 \cdot T' = T)$.

4-torsion

On the Kummer line, if $[T'] = (x : z)$ then:

$$(x : z) = [T'] = [-T'] = [3T'] = [T' + 2T'] = [T' + T] = (z : x)$$

2- and 4-torsion behaviour on the Kummer line

Let $E : y^2 = x(x^2 + \mathcal{A}x + 1)$, $T \in E$ a 2-torsion point and $P \in E$.

2-torsion

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$\rightsquigarrow \tau_T$, the translation by T is well-defined on the Kummer line.
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On the Kummer line, if $[T'] = (x : z)$ then:

$$(x : z) = [T'] = [-T'] = [3T'] = [T' + 2T'] = [T' + T] = (z : x)$$

We then solve $(x : z) = (z : x)$ iff $\frac{x}{z} = \pm 1$.
$\rightsquigarrow$ There is always a rational 4-torsion point $[T']$ above $[T]$ on the Kummer line.

Differential addition and doubling [Mon87]

Differential addition ($3M + 2S$)

Set $u := (x_P + z_P)(x_Q - z_Q)$ and $v := (x_P - z_P)(x_Q + z_Q)$

$$x_{P+Q} = (u+v)^2 \qquad z_{P+Q} = \frac{x_{P-Q}}{z_{P-Q}}(u-v)^2$$

Doubling ($2M + 2S + 1m_0$, $d = \frac{A+2}{4}$)

Set $u := (x_P + z_P)^2$, $v := (x_P - z_P)^2$ and $t := u - v$

$$x_{2\cdot P} = uv \qquad z_{2\cdot P} = t(v + dt)$$

$n = 22 = \overline{10110}^2$

P

Algorithm 1: Montgomery ladder step

Input: $[R] = [m \cdot P]$, $[S] = [(m+1) \cdot P]$, b a bit
Output: $([2 \cdot R], [R+S])$ if $b = 0$ $([R+S], [2 \cdot S])$ if $b = 1$
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    return ([R], [S]);
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Figure: Montgomery ladder

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$P \longrightarrow 2P$

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$n = 22 = \overline{10110}^2$

$$\begin{tikzcd} 0 & P \arrow[r] \arrow[d] \arrow[dr] & 2P \arrow[d] \\ & 2P \arrow[r, dash] & 3P \end{tikzcd}$$

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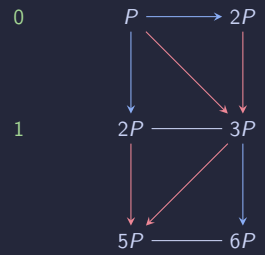

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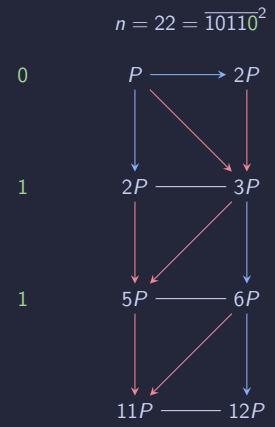

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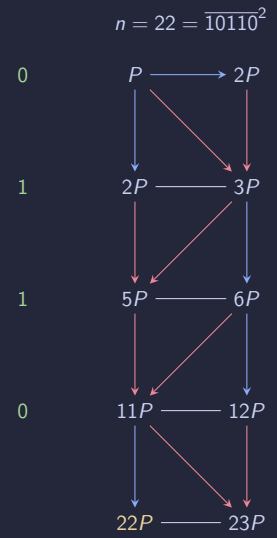

Figure: Montgomery ladder

Other coverings

Theta functions

Consider E an elliptic curve over $\mathbb{C}$. There exists analytic functions $(\theta_0(u), \theta_1(u))$ such that π is a degree 2 covering:

$$E \xrightarrow{\pi} \mathbb{P}^1$$
$$u \mapsto (\theta_0(u) : \theta_1(u))$$

In particular, $\pi^{-1}(\pi(P)) = \{\pm P\}$.

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In particular, $\pi^{-1}(\pi(P)) = \{\pm P\}$.

Theta constants: $a := \theta_0(0)$ and $b := \theta_1(0)$. They fully describe the map!
The image is denoted $\theta(a : b) \simeq \mathbb{P}^1$.

Even more models

Theta model (on E)

Constants: $a := \theta_0(0)$ and $b := \theta_1(0)$.
Notation: $\theta(a : b)$.

$$(x : z) := (\theta_0(u) : \theta_1(u)), u \in \mathbb{C}$$

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Notation: $\theta(a : b)$.

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Dual model (on E)

Constants: $a' := a + b$ and $b' := a - b$.
Notation: $\theta'(a' : b')$.

$$H : (x : z) \in \theta(a : b) \mapsto (x + z : x - z)$$

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Constants: a^2 and b^2.
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$$C : (x : z) \in \theta(a : b) \mapsto (ax : bz)$$

Theta squared (on E' w/ $E \xrightarrow{\text{2-isog}} E'$)

Constants: a^2 and b^2.
Notation: $\theta_s(a^2 : b^2)$

$$S : (x : z) \in \theta(a : b) \mapsto (x^2 : z^2)$$

Differential addition and doubling on $\theta(a:b)$

We get these from duplication formulas.
We set $2A'^2 := a^2 + b^2$ and $2B'^2 := a^2 - b^2$.

Differential addition ($3M + 4S + 1m_0$)

Set $u := (x_P^2 + z_P^2)(x_Q^2 + z_Q^2)$ and $v := \frac{A'^2}{B'^2}(x_P^2 - z_P^2)(x_Q^2 - z_Q^2)$

$$x_{P+Q} = u + v \qquad z_{P+Q} = \frac{x_{P-Q}}{z_{P-Q}}(u - v)$$

Doubling ($4S + 2m_0$)

Set $u := (x_P^2 + z_P^2)^2$ and $v := \frac{A'^2}{B'^2}(x_P^2 - z_P^2)^2$

$$x_{2.P} = u + v \qquad z_{2.P} = \frac{a}{b}(u - v)$$

Differential addition and doubling on $\theta_s(a^2 : b^2)$

Set $X = x^2$ and $Z = z^2$.
We still have the relation $2A'^2 = a^2 + b^2$ and $2B'^2 = a^2 - b^2$.

Differential addition ($3M + 2S + 1m_0$)

Set $u := (X_P + Z_P)(X_Q + Z_Q)$ and $v := \frac{A'^2}{B'^2}(X_P - Z_P)(X_Q - Z_Q)$

$$X_{P+Q} = (u+v)^2 \qquad Z_{P+Q} = \frac{X_{P-Q}}{Z_{P-Q}}(u-v)^2$$

Doubling ($4S + 2m_0$)

Set $u := (X_P + Z_P)^2$ and $v := \frac{A'^2}{B'^2}(X_P - Z_P)^2$

$$X_{2\cdot P} = (u+v)^2 \qquad Z_{2\cdot P} = \frac{a^2}{b^2}(u-v)^2$$

Comparison to Montgomery xz-coordinates

	Montgomery	Theta squared
Diff. Add.	$3M + 2S$	$3M + 2S + 1m_0$
Doubling	$2M + 2S + 1m_0$	$4S + 2m_0$

Table: Comparison of arithmetic

- Differential addition is always slower.
- With $S = \frac{2}{3}M$ and m_0 small, doubling should be faster.

Relations between Montgomery and theta squared (1/2) [HR19]

Consider the theta squared model $\theta_s(a^2 : b^2)$, and the Montgomery Kummer line M associated to:

$$E : y^2 = x\left(x - \frac{a^2}{b^2}\right)\left(x - \frac{b^2}{a^2}\right)$$

Isomorphism

There is an isomorphism between $\theta_s(a^2 : b^2)$ and M given by:

$$\theta_s(a^2 : b^2) \xrightarrow{\varphi} M$$

$$P_\theta = (x : z) \mapsto P_M = (a^2x - b^2z : b^2x - a^2z)$$

$$P_\theta = (a^2x - b^2z : b^2x - a^2z) \leftarrow\!\shortmid P_M = (x : z)$$

We have to spend some multiplications to convert...

Relations between Montgomery and theta squared (2/2) [KS20]

We use the same notations. These points are of 2-torsion on their respective models:

$$T_\theta = (1:0)\ (2 \cdot T_\theta = (a^2 : b^2)) \qquad T_M = (a^2 : b^2)\ (2 \cdot T_M = (1:0))$$

Translation map

There is a bijection between $\theta_s(a^2 : b^2)$ and M given by:

$$\theta_s(a^2 : b^2) \xrightarrow{\psi} M$$
$$P_\theta = (x : z) \mapsto P_M + T_M = (x : z)$$
$$P_\theta + T_\theta = (x : z) \leftarrow\!\shortmid P_M = (x : z)$$

Relations between Montgomery and theta squared (2/2) [KS20]

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We could then do something like this:

$$P_M \xmapsto{\psi^{-1}} P_\theta + T_\theta \xmapsto{[2]} 2 \cdot P_\theta \xmapsto{\psi} 2 \cdot P_M + T_M$$

Alternative formulas

On $E : y^2 = x\left(x - \frac{A_1}{B_1}\right)\left(x - \frac{B_1}{A_1}\right)$:

Differential addition (from Montgomery, $3M + 2S$)

Set $u := (x_P + z_P)(x_Q - z_Q)$ and $v := (x_P - z_P)(x_Q + z_Q)$

$$x_{P+Q} = (u+v)^2 \qquad z_{P+Q} = \frac{x_{P-Q}}{z_{P-Q}}(u-v)^2$$

Doubling up to $T := (A_1 : B_1)$ (from theta squared, $4S + 2m_0$)

Set $u := (x_P + z_P)^2$ and $v := \frac{A_2}{B_2}(x_P - z_P)^2$ where $A_2 := A_1 + B_1$, $B_2 := A_1 - B_1$

$$x_{2.P+T} = (u+v)^2 \qquad z_{2.P+T} = \frac{A_1}{B_1}(u-v)^2$$

Algorithm 2: Hybrid ladder step

Input: $[R]$, $[S]$ with $[R-S] \in \{[P],[P+T]\}$, b a bit

Output: $([2\cdot R+T],[R+S])$ if $b=0$

$([R+S],[2\cdot S+T])$ if $b=1$

Data: The points $[P]$, $[Q] := [P+T]$

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Function xDBLishADD([R], [S], b):
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    return ([R], [S]);
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$n = 22 = \overline{10110}^2$

P

Figure: Hybrid ladder

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$n = 22 = \overline{10110}^2$

$P \longrightarrow 2P + T$

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$n = 22 = \overline{10110}^2$

$$\begin{tikzcd}
0 & P \arrow[r] \arrow[d] \arrow[dr] & 2P + T \arrow[d] \\
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$$\begin{tikzcd}
0 & P \arrow[r] \arrow[d] \arrow[dr] & 2P + T \arrow[d] \\
1 & 2P + T \arrow[r, dash] \arrow[d] & 3P + T \arrow[d] \arrow[dl] \\
 & 5P \arrow[r, dash] & 6P + T
\end{tikzcd}$$

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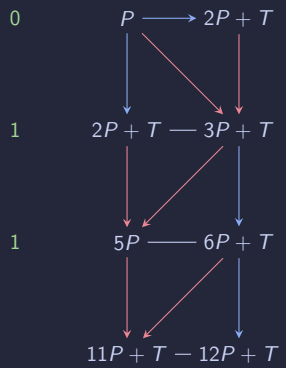

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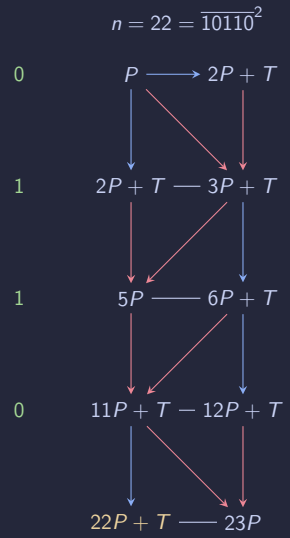

Figure: Hybrid ladder

Hybrid ladder proof of concept

Context

- $\mathbb{F}_{p^{10}} = \mathbb{F}_{p^5}[i]$ and $\mathbb{F}_{p^5} = \mathbb{F}_p[u]$ with $i^2 = -1$, $u^5 = 2$.
- Small multiplications are elements of $\mathbb{F}_p$ times elements of $\mathbb{F}_{p^{10}}$.
- Curve constants: $\alpha = 1 + \mu i$, $d = \frac{2-\alpha-\alpha^{-1}}{4} = \nu + i$, $\mu, \nu \in \mathbb{F}_p$
- 100 random scalar multiplications, repeated 100 times.

TL;DR small constants behave as small constants.

	Montgomery ladder	Hybrid ladder
Average (s)	2.502 ± 0.039	2.348 ± 0.017 (6.2%)

Table: Timings on Intel Core i5-1145G7 @ 2.60GHz

Future work and research direction

What's new?

- Hybrid ladder which provides better doubling (under specific circumstances).
- General framework to compute 2-isogenies on Kummer lines: new isogeny formulas, we find back the doubling formulas given earlier.

Work in progress

- Classification of elliptic curves given torsion properties on their Kummer line (via Galois representation).
- Choices of good curves for ECC and ECM to use hybrid ladder.

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